

课程名称	传感器与测试技术(双语)			考试日期	年 月 日	共 4 题
学生姓名		学院		班级		学号

Requirement: All the answers should be written on the answer sheet. And you can give the answers either in English or in 中文.

一、填空题 (每空 1 分, 共 20 分)

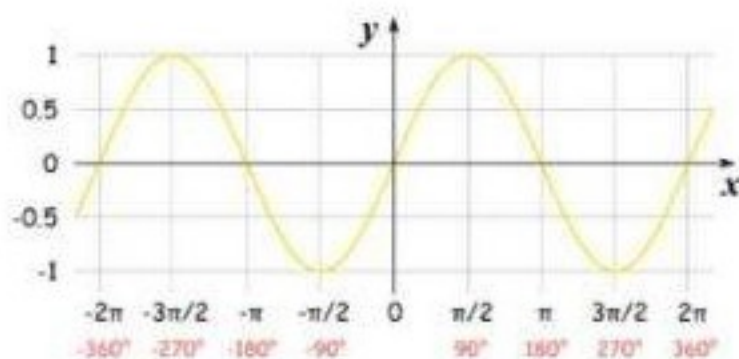
- In a measurement system, _____ takes the output of the sensing element and convert it into a form more suitable for further processing.
- For a periodic signal, its autocorrelation function is still a _____ signal and has the _____ frequency as that of the original periodic signal.
- The spectrum of periodic signal has three features, i.e., the spectrum is a _____ spectrum, the spectral lines appear only at the _____ and the _____, the amplitude of each components _____ with the increase in its frequency.
- The step response of a first-order instrument is exponential with _____, and _____, the faster the output approaches the inputs.
- Since the x is the input and y is output, and a, b, c, e, f and g is constant, we have $(a-e)\frac{dy}{dt} + by + (c-g) = (f-b)x$, and can get time constant τ in first-order instrument as $\tau =$ _____
- A sensor is a device which detects a signal or stimulus and produces a measurable output from it. A transducer is any device that transforms one form of energy to another. Hence a sensor is (usually) a _____. But not all transducers are strictly _____.
- The _____ is an effect in which energy is converted between mechanical and electrical forms.
- _____ is a device that allows a lower voltage to be derived from a higher voltage. The output is a scaled-down, or divided, version of the input.
- Capacitive displacement transducers produce a change in capacitance proportional to a change in displacement. The capacitance can be changed by varying _____, _____, and _____.
- The _____ of an element or a system is a range of frequencies for which $|G(j\omega)|$ is greater than $\frac{1}{\sqrt{2}}$.
- The purpose of signal matching using transformers is to ensure that _____ is transferred from the sensor to the display or recording device.
- _____ potentiometers measure angular displacement.

二、选择题（每题 2 分，共 10 分）

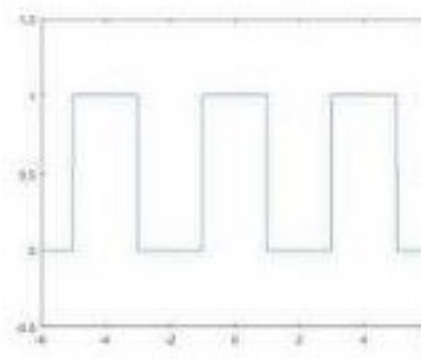
1. Please select the sensor used for measurement of motion. ()

- A. Sight glass B. Bimetallic thermostat
C. Microswitches D. Ultrasonic level indicator

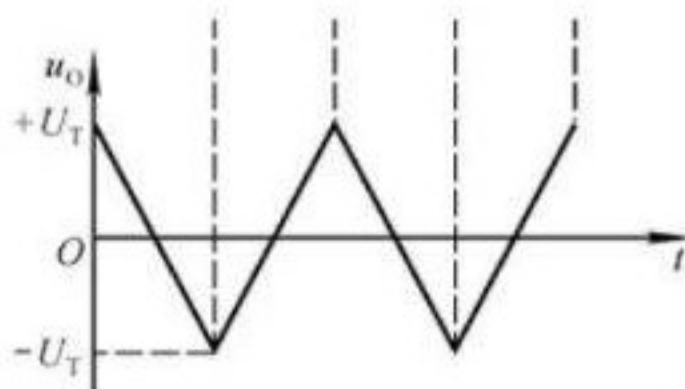
2. Which signal is not deterministic signal? ()



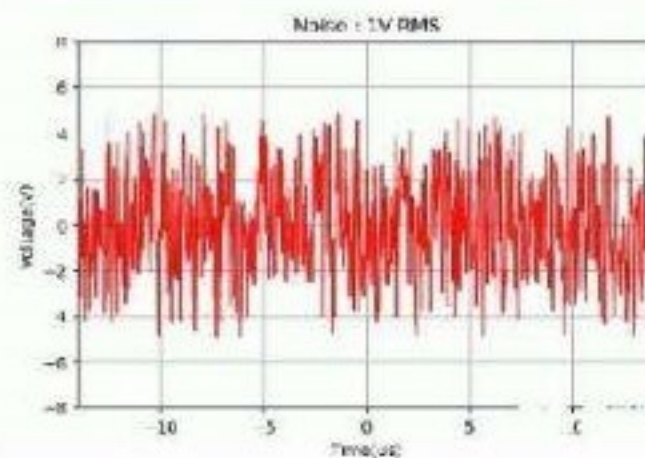
(A)



(B)



(C)



(D)

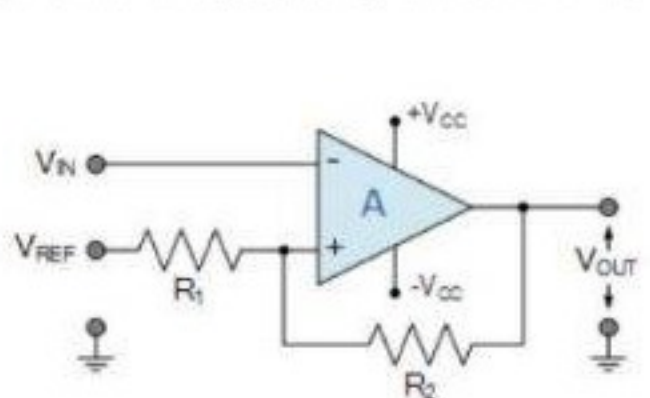
3. Which sensor can be used to measure the moisture content of sand? ()

- A. Hall sensor B. optical switch
C. Reed switch sensor D. Capacitance sensor

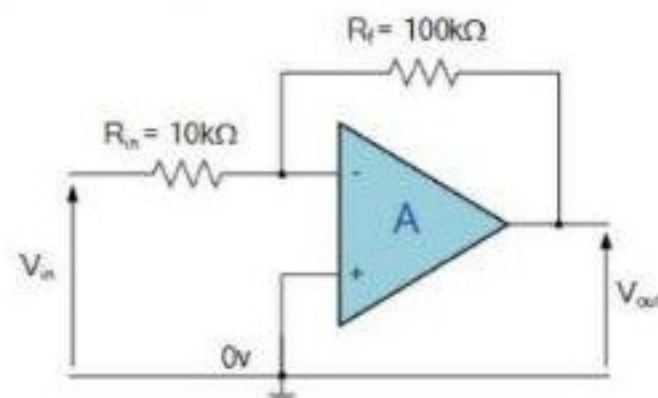
4. Which of the following sensors produces the minimum pressure loss? ()

- A. Nozzle B. Venturi tube C. Venturi nozzle D. Orifice plate

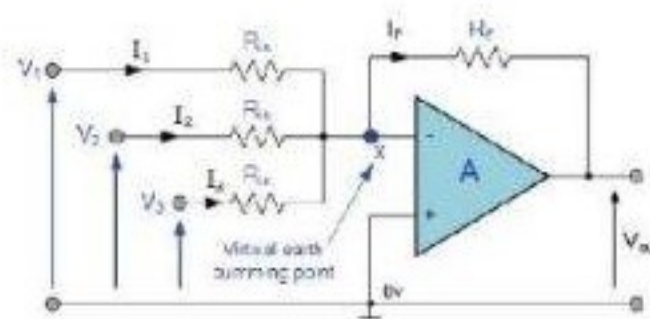
5. Which figure is voltage comparator circuit? ()



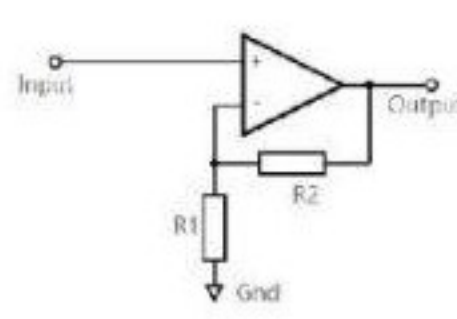
(a)



(b)



(c)



(d)

三、简答题（每题 10 分，共 40 分）

1. A time domain signal $F(t)$ and its frequency domain $F(\omega)$ are shown in Fig.1, and there is a typical cosine function $\cos\omega_0 t$ ($\omega_0 > \omega_m$). Please draw the amplitude chart of the FFT transform of $F(t) * \cos\omega_0 t$, and give the answer what will happen if $\omega_0 < \omega_m$.

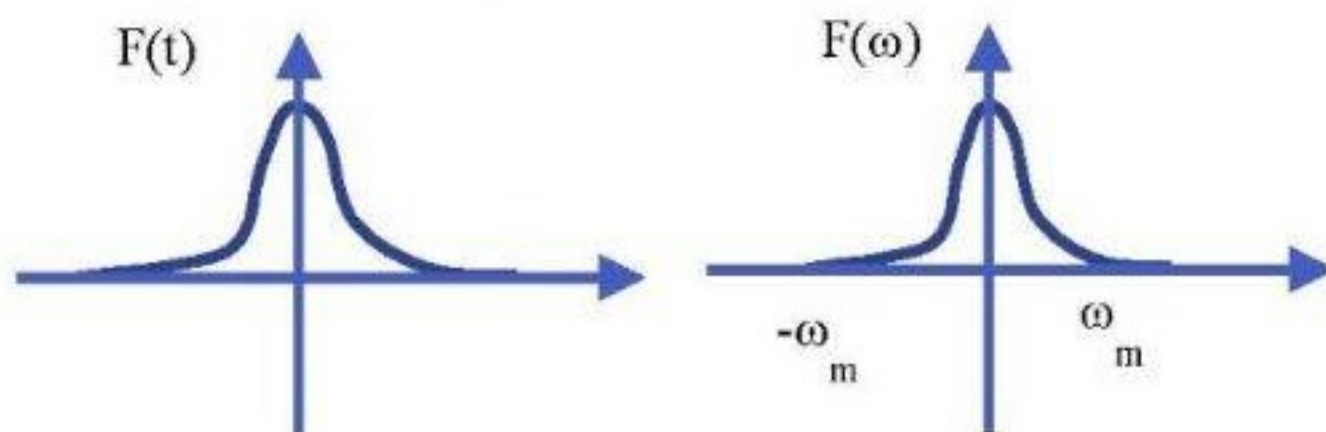


Fig.1 Time domain and its frequency domain chart

2. Please choose a type of sensor to detect the rotation speed of the gear, and explain its working principle.

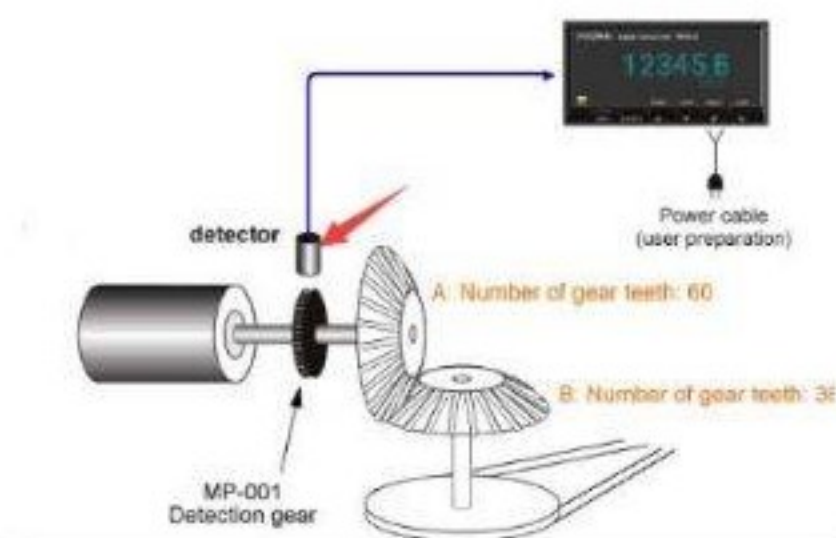


Fig.2 Measuring principle of rotation speed

3. Please explain the working principle of the sensor as shown in the Fig.3, and explain the phenomenon the rice cooker will be turn off when the cooking is done. (rice cooker 电饭煲)

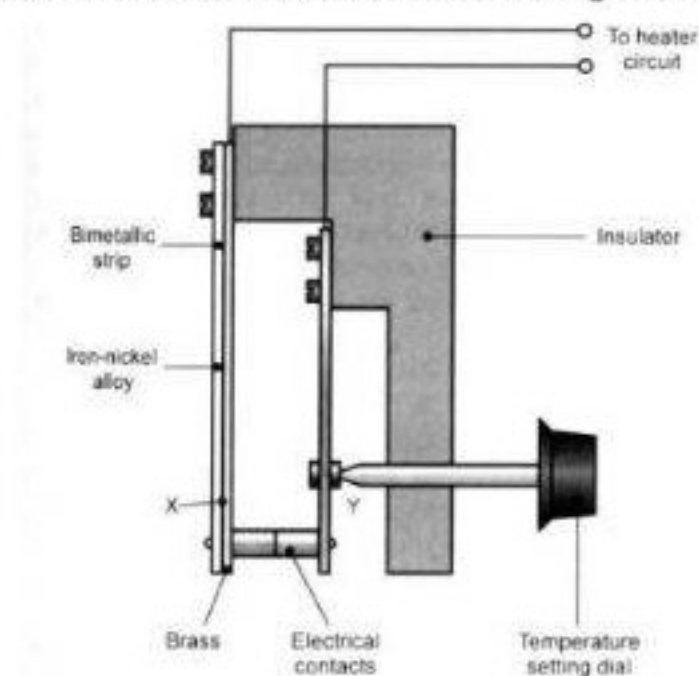


Fig.3

4. Please give a method to measure the fluid flow velocity and explain its working principle.

四、计算题（每题 10 分，共 30 分）

1. For a typical signal $x(t) = A \cos(\omega_0 t + \theta)$, $0 \leq t \leq T$, Please compute the mean value, mean square value, autocorrelation function and draw the frequency amplitude chart of $x(t)$. (Where $e^{jx} = \cos x + j \sin x$, $\cos^2 x = (1 + \cos 2x)/2$)
2. A low-pass first-order instrument has an amplitude inaccuracy of less than 5% when measuring a signal with frequency of 200Hz. What is maximal allowable time constant for the instrument? What is the phase angle at 50Hz and 100Hz?
3. As shown in Figure 5, the load P is applied to the end of the cantilever beam. The resistances of strain gauges R_1 and R_2 are equal to 120Ω . The gauge factor K of each strain gauge is 3. R_3 and R_4 with the resistance of 120Ω are the fixed resistors. The input voltage U_i is 12V and the strain ε at the fixed end of the cantilever beam produced by the load P is 200×10^{-6} .

(1) Plot the layout of gauges on cantilever beam and the sketch of gauges connected in a Wheatstone bridge to measure load P with highest bridge sensitivity.

(2) Calculate the change of resistance ΔR of each strain gauge.

(3) Determine the output voltage U_o .

(4) Can the temperature be compensated?

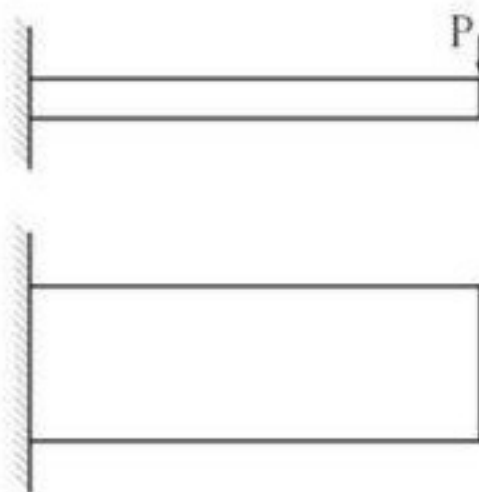


Fig.5 Cantilever beam